Flight Delays Are Contagious: Predicting Delay Propagation Across Consecutive Aircraft Operations

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***Abstract*—Aircraft delays can affect later operations when the same aircraft flies several legs in a day. This study examines whether the delay of a preceding flight helps predict the arrival delay of the next flight operated by the same aircraft. Using the 2025 U.S. Bureau of Transportation Statistics On-Time Performance data, we reconstruct aircraft rotations and obtain 6,768,487 valid consecutive-flight pairs. Features describe only the information available when the preceding aircraft reaches its destination gate. A classifier predicts whether the next flight arrives at least 15 minutes late, and a regressor estimates its arrival delay in minutes. Random Forest models were selected with a chronological training and validation split and evaluated on an independent December 2025 test set of 562,202 pairs. The classifier reaches a receiver operating characteristic area under the curve (ROC-AUC) of 0.7943 and an average precision of 0.7150; at an operating threshold of 0.35 it attains 72.52 percent precision and 53.24 percent recall. The regressor reaches a mean absolute error of 19.77 minutes and an R-squared value of 0.4851. Previous-flight delay, remaining turnaround time, and aircraft operational history therefore carry useful predictive information about subsequent flight delays. A shift-share decomposition attributes 94.0% to 98.4% of the rise in test error to the December delay mix, not model degradation.**

Keywords—aircraft operations, flight delay prediction, delay propagation, machine learning, Random Forest, turnaround time, aviation analytics

I. Introduction

Airlines assign each aircraft to a chain of flights during the day, so a late arrival can shorten the time available to prepare the aircraft for its next departure. This transfer of delay between consecutive operations is known as delay propagation [1]. Delays are also driven by airport congestion, weather, air-traffic restrictions, and airline scheduling, but the operating history of the aircraft is an additional and directly observable source of information: when an aircraft arrives late, the time left before its next scheduled departure may be too short to complete ground operations.

This study therefore treats consecutive flights of the same aircraft as linked observations rather than independent records. Aircraft rotations are reconstructed from tail numbers, airport continuity, and actual timestamps. The research question is: to what extent can delays from previous flight operations predict delays in subsequent flights operated by the same aircraft? The objectives are to (1) reconstruct consecutive aircraft operations from historical flight records, (2) quantify the association between preceding-flight and subsequent-flight delays, (3) develop classification and regression models using operational and aircraft-history features, and (4) evaluate the selected models on a chronologically separated, independent test set.

The work is a predictive decision-support analysis. It reports predictive association and does not claim that a preceding delay is the sole or direct cause of a subsequent delay.

II. Related Work

Delay propagation depends on schedule slack: AhmadBeygi et al. [2] and Pyrgiotis et al. [3] analyzed how delays spread through airline networks, and Lan et al. [4] showed that retimed departures and robust routing limit propagated disruption. Rebollo and Balakrishnan [5] used Random Forests on network-level delay states, and Gui et al. [6] applied machine learning to aviation big data. This study instead links consecutive legs of the same aircraft and tests on a later period.

III. Dataset and Preprocessing

A. Dataset Description

The study uses the U.S. Bureau of Transportation Statistics (BTS) On-Time Performance data for January through December 2025 [7], which provide scheduled and actual flight times, departure and arrival delays, carrier and airport details, and aircraft identifiers. The raw data contain 6,898,741 flight records from 14 airlines and 352 airports. They were processed with Python and DuckDB, with Parquet files used for intermediate and modeling datasets. Cancelled flights were not treated as completed movements, and diverted flights were kept as sequence-breaking events so that no rotation links flights across a diversion. Cancelled (104,748) and diverted (18,349) operations were isolated, and 110,996 further pairs with airport mismatches, diversion-induced breaks, missing predecessors, or invalid durations were removed. The final dataset contains 6,768,487 valid consecutive-flight pairs.

B. Aircraft Rotation Reconstruction

Consecutive operations were identified from tail numbers and actual operational timestamps converted to a common UTC timeline, accounting for airport time zones and daylight-saving transitions. A pair is valid when (i) both operations have a valid aircraft identifier, (ii) the destination of the preceding flight equals the origin of the subsequent flight, (iii) the preceding arrival occurs no later than the subsequent departure, (iv) the operations are in chronological order, and (v) no diverted operation lies between them. A subsequent flight is deliberately not excluded when its scheduled departure falls before the preceding flight's actual arrival, because such cases show a preceding flight that has consumed the scheduled turnaround time.

C. Prediction Point and Targets

The prediction point is the moment just after the preceding aircraft reaches its destination gate and before the

subsequent aircraft departs. Classification predicts whether the subsequent flight arrives at least 15 minutes late, following the BTS arrival-delay definition:

$$y = 1 \text{ if } d \geq 15, \text{ otherwise } y = 0 \quad (1)$$

where d is the arrival delay of the subsequent flight in minutes (NEXT_ARR_DELAY). Regression predicts d itself.

D. Feature Engineering

Three cumulative feature groups were built. Group A contains previous-flight and turnaround features: previous arrival and departure delays, their 15-minute delay indicators, the scheduled turnaround buffer, the estimated remaining turnaround time, and a turnaround-deficit indicator. Group B adds flight context: subsequent-flight carrier, origin, destination, distance, scheduled departure hour, day of the week, and month. Group C adds aircraft operational history: the previous same-day delayed-flight count, consecutive delayed-flight count, cumulative arrival delay, and historical delay rate. Only information available at the prediction point was used. The subsequent flight's actual departure delay, actual departure and arrival timestamps, actual ground and elapsed times, and retrospective BTS delay-cause fields were excluded to prevent information leakage, and history features were computed only from preceding operations, excluding the target flight.

E. Chronological Data Split

Pairs were split by the scheduled local operating date of the subsequent flight (Table I). The December 2025 test set was held out from model development and threshold selection; model and threshold choices relied on the training and validation periods only.

TABLE I. CHRONOLOGICAL SPLIT OF CONSECUTIVE-FLIGHT PAIRS

Dataset	Period (2025)	Flight pairs	Delayed flights
Training	Jan.–Sep.	5,065,376	1,111,200
Validation	Oct.–Nov.	1,140,909	230,521
Testing	Dec.	562,202	148,865
Total	**Jan.–Dec.**	**6,768,487**	**1,490,586**

IV. EXPLORATORY DATA ANALYSIS

A. Association Between Consecutive Delays

Across all 6,768,487 pairs, 22.02% of subsequent flights arrive at least 15 minutes late. The rate is 13.97% when the preceding flight is on time or early and 50.19% when it is delayed by at least 15 minutes (Table II and Fig. 1), a risk ratio of 3.59 and an odds ratio of 6.20. Preceding-flight status is therefore informative about subsequent delay risk, although the association may also reflect shared conditions such as airport congestion and schedule characteristics.

TABLE II. SUBSEQUENT-FLIGHT DELAY BY PRECEDING-FLIGHT STATUS

Preceding flight	Next on time	Next delayed	Delay rate
On time or early	4,528,318	735,329	13.97%
Delayed (≥15 min)	749,583	755,257	50.19%
Total	**5,277,901**	**1,490,586**	**22.02%**

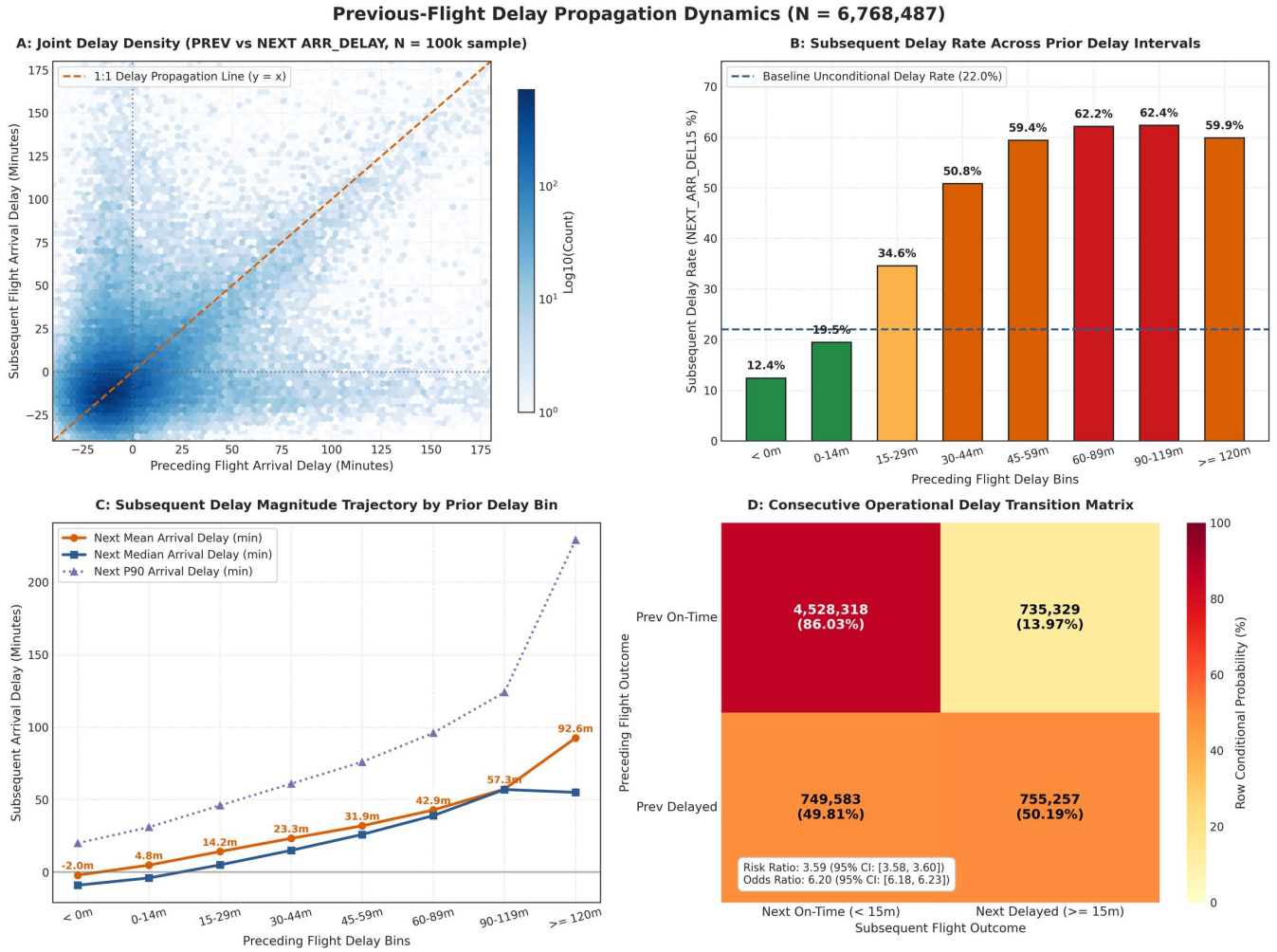


Fig. 1. Previous-flight delay propagation dynamics (N = 6,768,487): (A) joint density of preceding and subsequent arrival delays for a 100,000-pair sample; (B) subsequent delay rate by preceding-delay interval; (C) mean, median, and 90th-percentile subsequent arrival delay by preceding-delay interval; (D) transition matrix of consecutive delay outcomes with risk and odds ratios.

B. *Turnaround Buffer and Delay Propagation*

The scheduled turnaround buffer is the planned time between the preceding flight's scheduled arrival and the subsequent flight's scheduled departure. The estimated remaining turnaround time is this buffer reduced by the preceding arrival delay. When the arrival delay exceeds the buffer, the aircraft is in a turnaround deficit, which affects 382,937 pairs (5.66%). In the deficit cohort, 99.50% of subsequent flights are delayed, with a mean arrival delay of about 121.9 minutes; in the non-deficit cohort the figures are 17.38% and about 0.9 minutes (Fig. 2). Buffer exhaustion is therefore strongly predictive, although this association is not an isolated causal effect.

C. *Delay Severity and Class Imbalance*

Subsequent arrival delays are right-skewed: the median is −6 minutes, whereas the mean is about 7.78 minutes, so a small number of severe delays dominate the mean. The classification target is imbalanced (77.98% on time or early, 22.02% delayed), so accuracy alone is insufficient; precision, recall, F1-score, area under the receiver operating characteristic curve (ROC-AUC), and average precision (AP) are also reported.

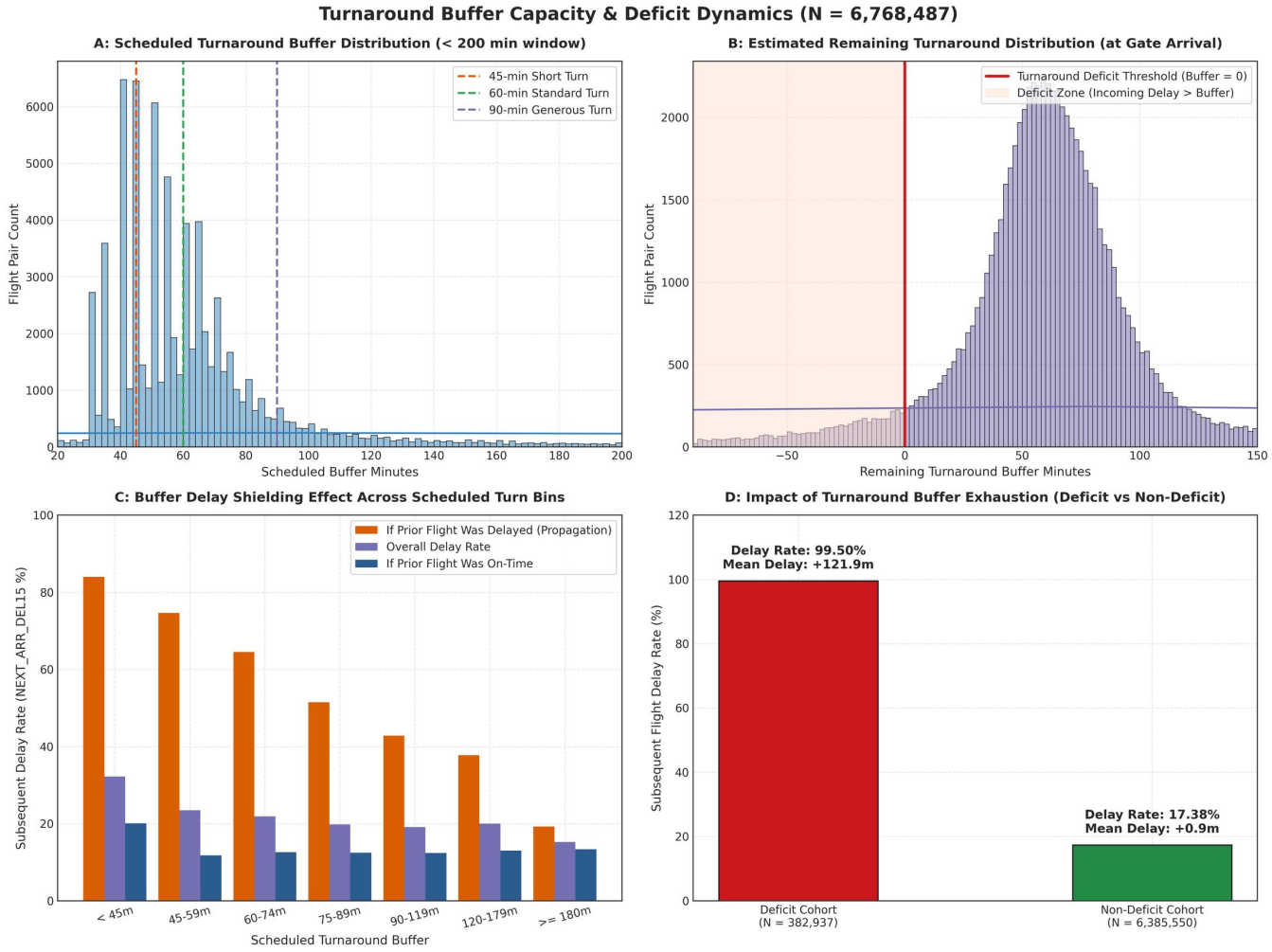


Fig. 2. Turnaround buffer capacity and deficit dynamics (N = 6,768,487): (A) scheduled turnaround buffer distribution; (B) estimated remaining turnaround time at gate arrival; (C) subsequent delay rate by scheduled buffer and preceding-flight status; (D) deficit versus non-deficit cohorts.

V. Model Development and Evaluation

A. *Classification Models*

Logistic Regression, Decision Tree, Random Forest, and Multi-Layer Perceptron classifiers were compared. The selected model is a Random Forest [8] trained on Group C features, which can represent nonlinear relationships between operational variables. It outputs a probability for the delayed class, and an operating threshold of 0.35 was fixed during validation-based model selection. This threshold gives a different precision-recall trade-off from the default of 0.50 and is not claimed to be globally optimal: on validation data a 0.01-step grid put the F1 maximum at 0.39 (F1 = 0.5751), the default of 0.50 gave 86.68% precision but only 40.88% recall, and 0.35 gave 69.20% precision, 49.03% recall, and F1 = 0.5740 and was fixed before test evaluation. Models were implemented with scikit-learn [9].

B. *Regression Models*

Ordinary least squares, Decision Tree, Random Forest, and Multi-Layer Perceptron regressors were compared. The selected model is a Random Forest regressor trained on Group C features, which captures nonlinear interactions among turnaround information, previous-flight delays, flight

context, and aircraft operational history. Linear models used all 5,065,376 training pairs, tree models a stratified sample of 500,000 pairs, and the neural networks 250,000, all without regularization.

C. Evaluation Metrics

Classification is evaluated with accuracy, precision, recall, F1-score, ROC-AUC, and AP, which summarizes the precision-recall curve and is informative under class imbalance [10]. Regression is evaluated with mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R-squared). RMSE weights large errors more heavily, and R-squared is measured against a mean-prediction baseline.

D. Permutation Importance

Permutation importance measures how much a fitted model relies on a feature: the decrease in AP for the classifier and the increase in MAE for the regressor after the feature is shuffled. It indicates model reliance, not causal effects, and can be distorted by correlated or derived variables.

VI. Experimental Results

A. Classification Performance

On the December 2025 test set (562,202 pairs, of which 148,865 are delayed and 413,337 are on time or early), the classifier achieves a ROC-AUC of 0.7943 and an AP of 0.7150, well above the 26.48% positive-class prevalence (Table III, Figs. 3 and 4). At the 0.35 threshold it attains 72.52% precision, 53.24% recall, and an F1-score of 0.6140.

TABLE III. Classification Performance on the December 2025 Test Set

Metric	Result	Metric	Result
Test observations	562,202	Accuracy	82.28%
Positive prevalence	26.48%	Precision	72.52%
ROC-AUC	0.7943	Recall	53.24%
Average precision	0.7150	F1-score	0.6140
Threshold	0.35	Specificity	92.74%

The confusion matrix is given in Table IV. The model correctly flags 79,250 delayed flights, raises 30,025 false alerts, and misses 69,615 delayed flights, so it identifies slightly more than half of all delayed flights while keeping precision above 70%. At a threshold of 0.50, precision rises to 87.52% and recall falls to 45.07%; the threshold of 0.35 is therefore a higher-coverage operating choice rather than a claim of universal optimality.

TABLE IV. Confusion Matrix at Threshold 0.35

Actual class	Predicted on time	Predicted delayed
On time or early	383,312	30,025
Delayed (≥15 min)	69,615	79,250

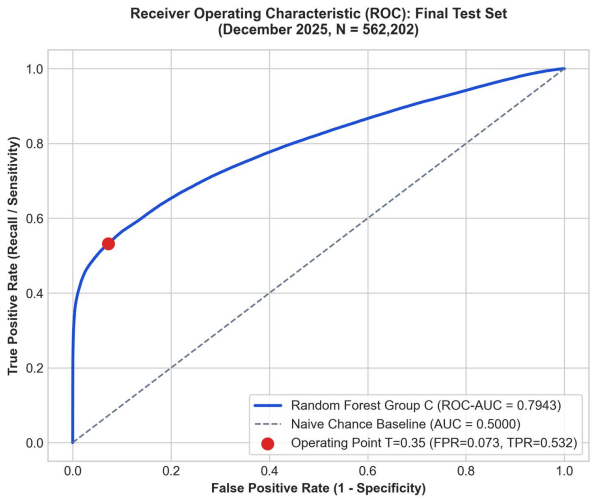


Fig. 3. Receiver operating characteristic curve of the Random Forest classifier on the December 2025 test set (ROC-AUC = 0.7943). The marker shows the 0.35 operating point.

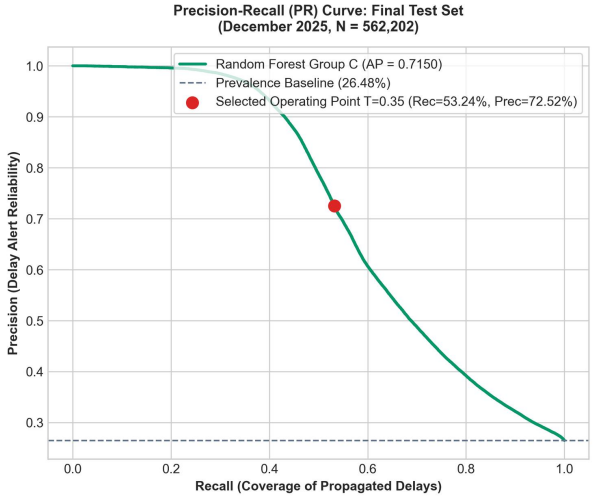


Fig. 4. Precision-recall curve of the Random Forest classifier on the December 2025 test set (AP = 0.7150) against the 26.48% prevalence baseline. The marker shows the 0.35 operating point.

B. Regression Performance

On the same test pairs, the Random Forest regressor reaches an MAE of 19.77 minutes, an RMSE of 45.86 minutes, and an R-squared of 0.4851 (Table V), explaining about 48.5% of the variance in subsequent arrival delay relative to a mean-prediction baseline. Fig. 5 compares predicted and actual delays for a 50,000-pair sample: predictions concentrate in the central delay range, while large actual delays deviate from the identity line, showing that the magnitude of extreme delays is hard to estimate.

TABLE V. Regression Performance on the December 2025 Test Set

Pairs	MAE (min)	RMSE (min)	MSE (min²)	R-squared
562,202	19.77	45.86	2,103.41	0.4851

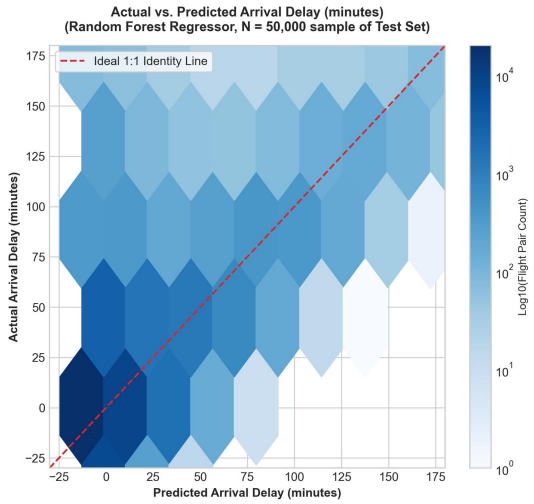


Fig. 5. Actual versus predicted subsequent arrival delay for a 50,000-pair sample of the December 2025 test set (hexagonal bin counts, logarithmic color scale). Axes are limited to −30 to 180 minutes, and the dashed line marks ideal one-to-one prediction.

C. Error Across Delay Severity Bands

Test pairs were grouped into five bands by actual delay (Table VI and Fig. 6). Errors are smallest for early, on-time, and minor delays and rise sharply for severe delays (MAE 109.98 minutes, RMSE 201.18 minutes), where a small number of very large errors dominate the squared error and the mean prediction (140.63 minutes) is far below the actual mean (231.01 minutes). Bias is negative for the moderate, significant, and severe bands, indicating systematic underprediction of larger delays, and positive (+12.27 minutes) for the early or on-time band.

TABLE VI. REGRESSION ERROR ACROSS DELAY SEVERITY BANDS

Actual delay band	Pairs	MAE (min)	RMSE (min)	Bias (min)
Early/on time (≤0)	322,255	13.01	16.69	+12.27
Minor (1–14)	91,082	10.58	12.64	−3.46
Moderate (15–59)	94,233	22.27	26.57	−13.64
Significant (60–119)	31,939	42.70	54.03	−30.78
Severe (≥120)	22,693	109.98	201.18	−90.38

Bias is the mean of predicted minus actual delay, in minutes.

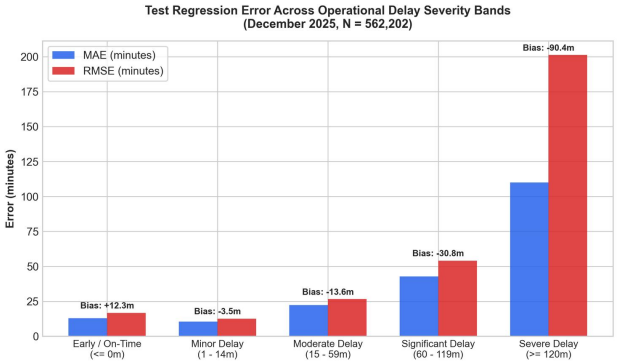


Fig. 6. MAE and RMSE of the regressor across five delay-severity bands on the December 2025 test set (N = 562,202). Annotations show the mean bias in minutes.

D. Test-Set Permutation Importance

Importance was computed on a 100,000-pair sample of the test set with five repeats (Table VII and Fig. 7). For the classifier, estimated remaining turnaround time causes the largest AP decrease, followed by the turnaround-deficit indicator and the scheduled turnaround buffer; previous arrival delay and aircraft consecutive-delay history also contribute. For the regressor, remaining turnaround time dominates (a 15.81-minute MAE increase), and the other features have far smaller effects. Because remaining turnaround time is derived from the scheduled buffer and previous arrival delay, permuting it alone breaks that relationship; its importance reflects model reliance under perturbation, not an isolated causal contribution; permuting it drops the regressor's R-squared on the sample from 0.4770 to −0.4415.

TABLE VII. LEADING FEATURES IN TEST-SET PERMUTATION IMPORTANCE

Feature	Classifier: mean AP decrease	Regressor: mean MAE increase (min)
Remaining turnaround time	0.0303	15.81
Turnaround-deficit indicator	0.0219	≈0
Scheduled turnaround buffer	0.0129	0.48
Previous arrival delay	0.0105	0.93
Aircraft consecutive delayed flights	0.0082	0.84
Next scheduled departure hour	≈0.004	≈0.90

Values marked ≈ are read from Fig. 7.

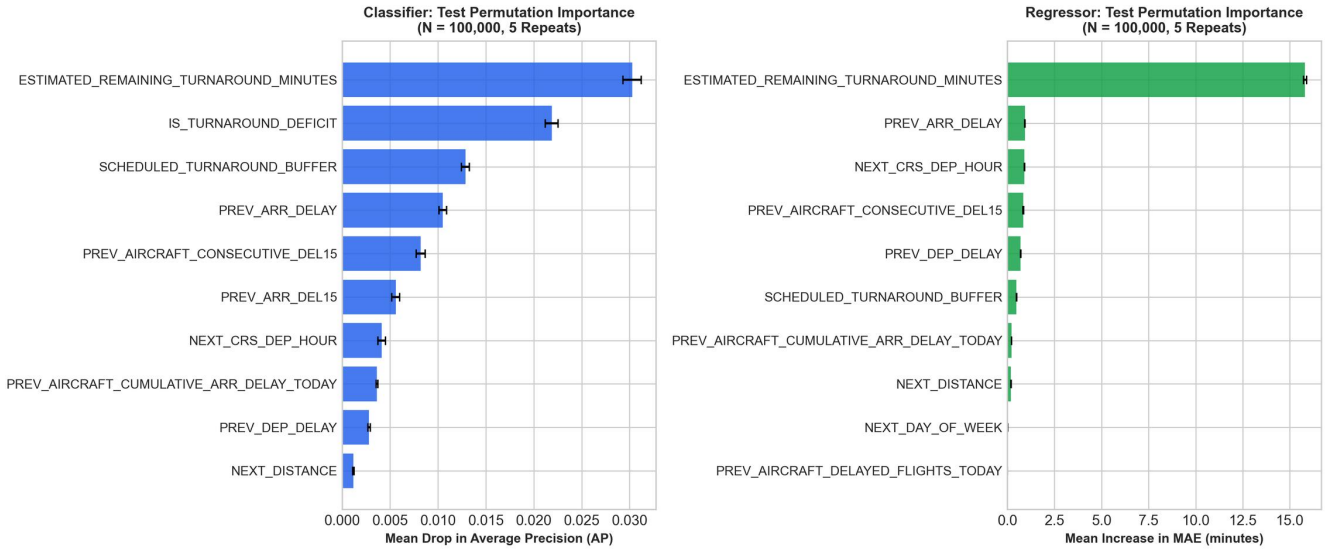


Fig. 7. Test-set permutation importance of the Random Forest classifier (mean decrease in AP) and regressor (mean increase in MAE, in minutes), computed on 100,000 test pairs with five repeats. Error bars show variation across repeats.

E. Validation-Set Model Comparison

Table VIII shows the Random Forest with Group C features discriminating best on validation data, with an MAE of 17.40 minutes against 21.19 for ordinary least squares. Adding flight context (Group A to B) raised Logistic Regression ROC-AUC by 0.0341 and same-day aircraft history (B to C) by a further 0.0077; for the Random Forest, Group C improved on Group A by 0.0108.

TABLE VIII. VALIDATION-SET COMPARISON OF MODEL FAMILIES

Model	Group	ROC-AUC	AP	MAE (min)	R^2
Baseline	—	0.5000	0.2021	22.65	−0.0479
Logistic / OLS	C	0.7775	0.6228	21.19	0.2754
Decision Tree	C	0.7656	0.5983	17.89	0.4281
MLP	C	0.7818	0.6295	17.06	0.4304
Random Forest	A	0.7824	0.6433	17.58	0.4440
Random Forest	C	0.7932	0.6600	17.40	0.4516

F. Seasonal Shift and Error Decomposition

From validation to test, ROC-AUC stayed at 0.7932 and 0.7943 while AP rose from 0.6600 to 0.7150 with prevalence (20.21% to 26.48%), but MAE rose from 17.40 to 19.77 minutes and RMSE from 38.62 to 45.86. To test whether this reflects degradation or a change in delay mix, the MAE change was decomposed over the bands of Table VI [11], [12]:

$$\Delta\text{MAE} = \Sigma\, w_{t,i}\,(e_{t,i} - e_{v,i}) + \Sigma\,(w_{t,i} - w_{v,i})\, e_{v,i} \quad (2)$$

where w and e are the share and MAE of band i in the test (t) and validation (v) periods. With test weights, the within-band effect is +0.038 minutes (1.59%) and the composition effect +2.330 minutes (98.41%); with symmetric Shapley weights the composition effect is 93.99% (+2.225 minutes). In the moderate band, MAE fell from 22.44 to 22.27 minutes. The higher error therefore reflects a 55.4% surge in severe delays (≥120 minutes) in winter rather than model decay, although one validation–test pair gives only descriptive evidence.

VII. DISCUSSION

The results support the use of preceding-flight information for predicting subsequent delays. The subsequent delay rate rises from 13.97% after an on-time or early preceding flight to 50.19% after a delayed one, and from 17.38% outside the turnaround-deficit cohort to 99.50% inside it, which marks the remaining turnaround time as a strong indicator of operational risk.

The classifier ranks delayed and non-delayed pairs usefully (ROC-AUC 0.7943, AP 0.7150, against a 26.48% prevalence baseline). At the 0.35 threshold, 72.52% precision with 53.24% recall may suit an alerting setting in which finding a meaningful share of at-risk flights matters. The regressor explains 48.51% of delay variance, but its error grows with severity and it underestimates large delays, so it is better suited to ranking risk than to forecasting the length of severe disruptions.

Permutation analysis identifies remaining turnaround time as the most influential feature for both tasks, with aircraft history also contributing to classification. Because several turnaround variables are mathematically or operationally related, these results describe model dependence rather than independent causal effects. The findings are evidence that aircraft rotation information has predictive value, not proof that the preceding flight is the only source of a subsequent delay.

VIII. LIMITATIONS AND FUTURE WORK

The study has seven limitations. First, it uses 2025 U.S. domestic operations only, so results may not generalize to international operations, other years, or networks with different scheduling practices. Second, features are restricted to information available at the prediction point; real-time weather, traffic-management decisions, maintenance events, and gate availability are not comprehensively represented and may explain part of the remaining error. Third, severe delays are rare, which weakens regression performance for extreme delays. Fourth, permutation importance can be affected by correlated and derived variables, so the importance of remaining turnaround time is not a causal estimate. Fifth, a single December holdout gives only one

temporal test period; evaluation across several periods and seasons would give stronger evidence of generalization. Sixth, buffer deficits predicting delay does not show that added padding would remove downstream delay one for one, because carriers adjust operations to schedule buffers. Finally, Random Forest predictions are averages over terminal leaves (minimum leaf size 50), so the regressor cannot anticipate extreme multi-hour disruptions that develop after departure.

Future work can add real-time weather, airport congestion, air-traffic flow restrictions, and maintenance data, and can examine probability calibration, cost-sensitive alert thresholds, and performance across airlines, airports, and aircraft types.

IX. Conclusion

This study asked whether the delay of a preceding flight helps predict the arrival delay of the next flight operated by the same aircraft. Using 6,768,487 consecutive-flight pairs reconstructed from the 2025 BTS On-Time Performance data, the subsequent delay rate rose from 13.97% after an on-time or early preceding flight to 50.19% after a delayed one, and turnaround-deficit operations showed a 99.50% subsequent delay rate.

On the independent December 2025 test set, the Random Forest classifier achieved a ROC-AUC of 0.7943 and an AP of 0.7150, with 72.52% precision and 53.24% recall at a threshold of 0.35, and the Random Forest regressor achieved an MAE of 19.77 minutes and an R-squared of 0.4851. Previous-flight delays, turnaround availability, and aircraft operational history thus provide useful information for estimating subsequent delay risk and can support predictive monitoring of aircraft rotations. A shift-share decomposition attributes 94.0% to 98.4% of the rise in test MAE to the heavier December delay mix rather than to model decay, and the turnaround buffer is the most influential predictor.

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